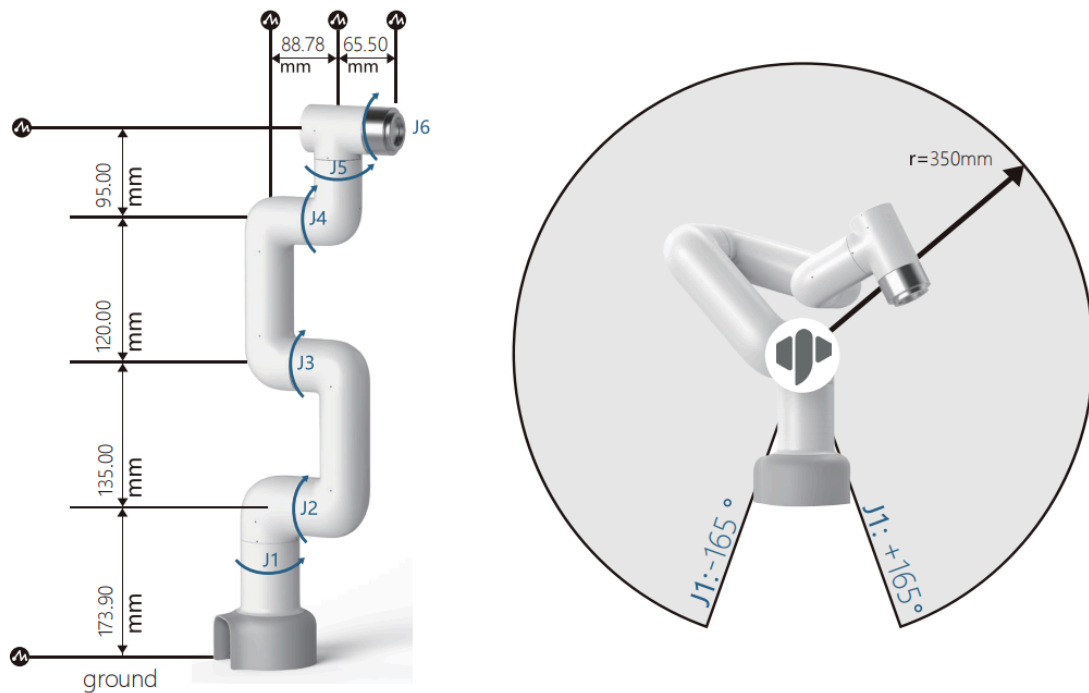


Coordinate System

1 Joint coordinate system

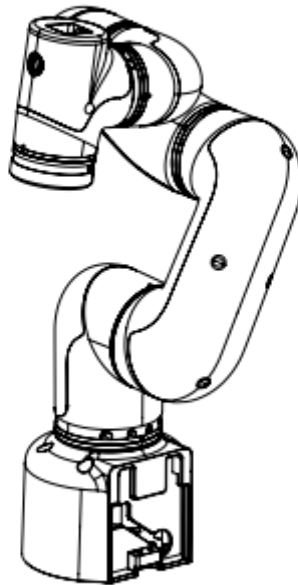
The joint coordinate system is determined based on each moving joint. All joints are revolute joints. The range of motion is as follows



joint	scope
J1	-168 ~ +168
J2	-135 ~ +135
J3	-145 ~ +145
J4	-148 ~ +148
J5	-168 ~ +168
J6	-180 ~ +180

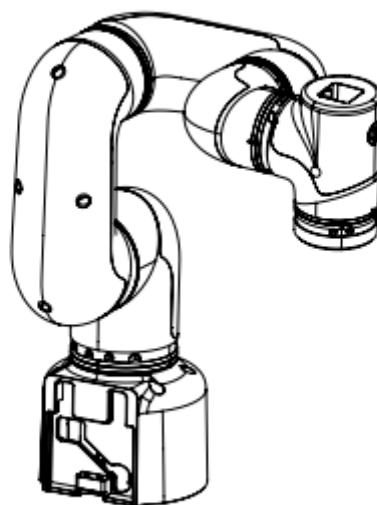
2 User coordinate system

The user coordinate system is a user-defined workbench coordinate system or workpiece coordinate system. Its origin and axis direction can be determined according to actual needs, which is convenient for measuring the position of each point in the working area and arranging tasks. The default user coordinate system is determined based on the center point of the robot base, and the positive direction of the Y axis is the direction of the heavy load line.

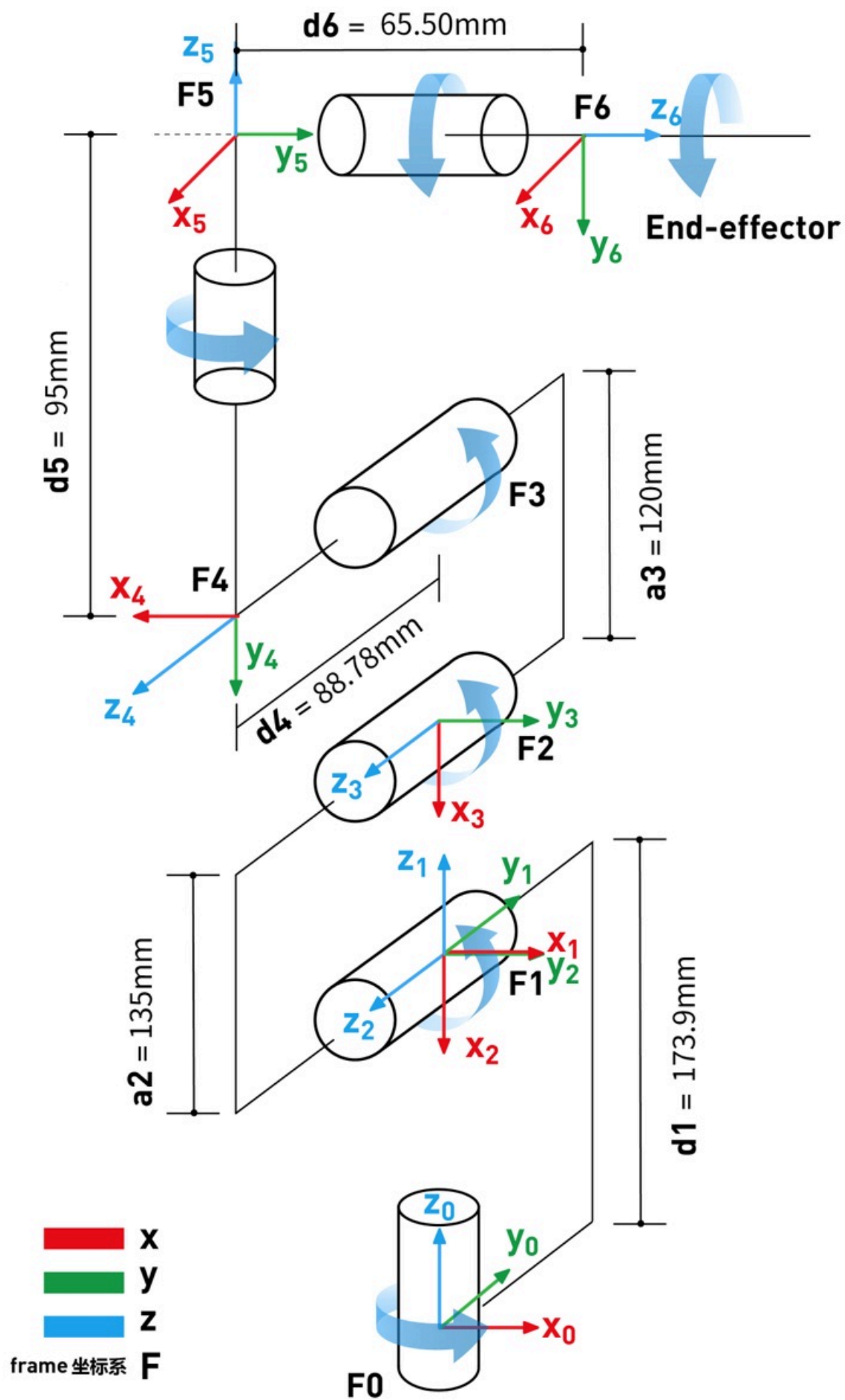


3 Tool coordinate system

The tool coordinate system is a coordinate system that defines the position of the tool center point (TCP) and the tool posture. Its origin and direction change with the position and angle of the final workpiece. The default tool coordinate system is determined based on the center point of the tool flange, and the positive direction of the Y axis is opposite to the direction of the aviation socket.



4 Joint connecting rod parameters and specifications



4.1 DH Range

For the rotation joint n , set $\theta = 0.0$, where the X axis is consistent with the X axis direction, and select the origin position of the coordinate system (N) to satisfy $d = 0.0$. For the translation joint n , set the direction of the Z axis to satisfy $\theta = 0.0$. When $d = 0.0$, select the origin of the coordinate system (N) at the intersection of the X_{N-1} axis and the joint axis n .

Summary of linkage parameters in linkage coordinate system If the linkage coordinate system is fixed on the linkage device according to the above provisions, the linkage parameters can be defined as follows:

- a_{i-1} : along x_{i-1} : the distance from z_{i-1} to z_i
- α_{i-1} : around x_{i-1} : from z_{i-1} to z_i
- d_i : represents the distance from x_{i-1} to x_i along z_i
- θ_i : around z_i : the angle from x_{i-1} to x_i

Here is an article to refer to <https://blog.csdn.net/hitgavin/article/details/104442034>

4.2 SDH parameter list

joint	theta	d	alpha	a	offset
1	theta_1	173.9	PI/2	0	0
2	theta_2	0	0	-135	-PI/2
3	theta_3	0	0	-120	0
4	theta_4	88.78	PI/2	0	-PI/2
5	theta_5	95	-PI/2	0	0
6	theta_6	65.5	0	0	0